

TYRA OEST, Denmark

WMO: 060220

Lat: 55.721N

Lon: 4.802E

Elev: 246

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.6	32.3	20.8	15.7	33.2	22.7	17.3	34.0	56.2	41.3	52.5	41.4	25.5	100	1.040

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	66.3	63.1	64.7	62.0	64.2	61.6	63.9	65.3	62.8	64.5	61.8	63.6	17.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.8	64.3	62.3	84.8	63.9	60.9	80.5	63.1	29.3	66.1	28.4	64.0	27.6	63.6	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 51.2	46.8	42.6	DB	30.2	67.4	2.2	2.0	28.6	68.8	27.3	70.0	26.1	71.2	24.5	72.6	(4)
(5)			WB	26.5	64.5	2.4	1.2	24.7	65.3	23.3	66.0	21.9	66.7	20.1	67.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	41.3	40.3	41.8	44.4	49.3	54.8	59.6	61.1	58.5	54.0	48.2	43.7
	DBStd	8.05	3.90	3.76	3.34	3.23	3.73	3.11	2.98	2.55	2.89	3.25	4.26	3.80
	HDD50	1306	270	272	254	170	57	3	0	0	0	7	78	195
	HDD65	5547	735	692	719	618	486	305	169	124	194	342	503	660
	CDD50	1237	0	0	0	1	36	148	298	343	256	130	25	0
	CDD65	4	0	0	0	0	0	0	2	2	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	22.6	27.5	24.8	23.1	20.0	19.9	18.6	18.5	19.6	21.7	24.6	26.6	26.0
Precipitation	PrecAvg	44.6	5.5	4.4	3.7	2.2	1.8	2.4	2.8	3.5	3.9	4.3	5.1	5.1
	PrecMax	55.3	9.6	7.3	6.5	3.9	2.7	4.3	4.0	5.7	6.5	6.2	8.1	9.9
	PrecMin	36.1	1.9	1.9	1.3	0.9	0.9	0.9	1.3	1.5	1.8	1.7	3.2	2.5
	PrecStd	4.4	1.9	1.5	1.2	0.7	0.5	0.7	0.8	0.9	1.1	1.2	1.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	46.8	48.3	53.2	60.9	64.4	68.2	68.2	66.3	61.1	57.5	52.0
		MCWB	47.8	45.1	45.9	48.7	56.8	60.9	63.8	64.8	63.8	59.0	55.2	50.3
	2%	DB	48.4	46.6	46.8	51.4	58.7	62.4	66.3	66.4	64.5	59.4	55.8	50.4
		MCWB	46.6	44.9	45.0	47.7	55.6	59.7	62.7	63.4	61.9	57.1	53.4	48.6
	5%	DB	46.8	46.0	46.3	49.9	55.8	60.7	64.7	64.8	63.0	59.1	55.0	49.9
		MCWB	45.1	44.5	44.7	47.5	52.9	58.3	62.1	62.0	60.3	56.7	52.4	48.1
	10%	DB	46.4	44.8	45.0	48.4	53.8	59.0	64.0	64.4	62.5	58.6	53.6	48.5
		MCWB	44.6	43.4	43.4	46.9	51.1	56.8	61.6	61.6	59.8	56.1	50.9	46.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	45.9	45.9	50.3	59.1	61.5	64.5	66.3	63.9	59.8	56.3	51.1
		MCDB	48.6	46.3	46.6	51.5	59.8	63.2	66.6	67.6	65.7	61.0	57.3	51.7
	2%	WB	46.7	45.0	45.1	48.6	55.6	59.9	63.4	64.5	62.4	58.5	54.6	49.5
		MCDB	47.5	45.6	45.9	49.8	58.0	61.8	65.4	65.4	64.1	59.7	56.2	50.0
	5%	WB	45.6	44.2	44.5	47.6	53.6	58.7	62.2	63.2	61.3	57.7	52.8	48.1
		MCDB	46.6	44.9	45.3	48.9	56.3	60.5	64.2	64.6	63.0	59.0	54.6	48.7
	10%	WB	44.6	43.4	43.8	46.6	52.0	57.2	61.5	62.0	60.1	56.8	51.5	47.0
		MCDB	45.7	44.3	44.7	48.0	53.9	59.1	63.4	63.8	61.9	58.2	53.4	48.0
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.2	3.3	3.3	3.5	3.2	3.2	3.0	3.0	3.3	3.8	3.7
		MCDBR	4.2	3.3	3.5	4.1	6.1	4.9	4.4	3.6	3.1	3.3	4.1	4.2
		MCWBR	5.0	3.3	3.2	3.0	4.6	3.5	3.2	3.7	3.7	4.0	5.1	5.2
	5% WB	MCDBR	3.9	3.0	3.3	3.8	6.2	4.2	4.0	3.5	2.7	3.4	3.8	3.5
		MCWBR	4.9	3.5	3.4	3.2	4.6	3.5	3.3	4.3	3.6	4.2	5.2	4.7
Clear-Sky Solar Irradiance	taub	0.332	0.345	0.364	0.396	0.378	0.384	0.388	0.390	0.376	0.355	0.334	0.312	
	taud	2.228	2.312	2.324	2.283	2.414	2.415	2.415	2.415	2.433	2.391	2.286	2.227	
	Ebn at Noon	177	227	253	261	274	273	269	261	247	221	176	154	
	Edh at Noon	18	24	30	36	34	34	34	32	27	23	17	15	
All-Sky Solar Radiation	RadAvg	146	350	787	1330	1779	1809	1718	1412	920	473	190	105	
	RadStd	20	55	75	120	137	131	168	108	102	44	25	14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page